

Carlos Loya

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PROFESSIONAL SUMMARY

Software developer with 3 years of professional experience in Go, Python, TypeScript, and Kubernetes, plus 3 years of independent project work delivering production-grade systems for real clients. Comfortable across the full stack: backend services, data pipelines, infrastructure, and React frontends. Recent work spans a Kubernetes operator packaged as a Helm chart, a Go-based water quality compliance platform, and a multi-sport ETL pipeline currently in production use.

SKILLS

Languages: Go, Python, TypeScript, JavaScript, Bash, SQL

Backend & Infrastructure: Kubernetes, Docker, Helm, PostgreSQL, TimescaleDB, Kafka, AWS (EC2, S3, ECS), Terraform

Frontend: React, Next.js, Vue.js, Tailwind CSS

Data: Apache Airflow, DuckDB, Parquet, Medallion architecture

AI Tools: GitHub Copilot, Claude, Cursor

EXPERIENCE

Independent Software Developer

Nov 2022 – Present

McAllen, TX (Remote)

- Worked on independent software projects throughout, from small weekend builds to production-grade client work. Larger deliverables include a Kubernetes operator, a Go-based water quality compliance platform, a multi-sport ETL pipeline, and a full-stack lifestyle blog.

Software Developer | InfluxData

Sept 2021 – Nov 2022

Remote, USA

- Wrote a Go microservice that backs up Kafka messages to cloud object stores (S3 and others), covering disaster recovery, event replay, debugging, and compliance use cases.
- Set up Kafka rack awareness at the cluster level (broker configs + partition assignments), which cut inter-region traffic and took 13% off the team's cloud bill.

Software Developer | Comcast

Nov 2019 – Sept 2021

Austin, TX

- Contributed Go modules, bug fixes, and support tooling to Plax, an open-source test framework for Xfinity's IoT platform. Wrote the original Bash script that eventually became the plaxrun CLI.
- Managed Concourse CI pipelines and Terraform infrastructure for several teams, covering testing and deployments to AWS ECS and Comcast's Cloud Foundry platform.

PROJECTS

Water Quality Compliance Platform

Mar 2026

- Built a water quality compliance platform in Go with a React/TypeScript frontend, modeled after a real city RFI. Multi-tenant schema covers organizations, facilities, monitoring locations, and permit limits, with PostgreSQL EXCLUDE constraints preventing date range overlaps.
- Every data change fires a NATS JetStream event written to an append-only TimescaleDB audit log. Compliance engine checks each sample against the permit limits active at collection time. Built Excel and PDF exports with exceedance row highlighting.

Archive Purge Restore – Kubernetes Operator & CLI

Feb 2026

- Built a Kubernetes operator for database row archival packaged as a Helm chart with CRDs and validating admission webhooks. Supports PostgreSQL, MySQL, and TimescaleDB with pluggable S3, GCS, R2, and local storage backends.
- Two-phase archive algorithm writes rows to a pending Parquet file before deleting from the source database so no data is lost if the upload fails. Ships with Prometheus metrics, a Grafana dashboard, and a kind end-to-end test suite.

Sports Data Pipeline – Multi-Sport ETL

Mar 2026

- Built a sports analytics pipeline covering NBA, NFL, and soccer data through a Bronze to Silver to Gold medallion architecture, landing in DuckDB tables with rolling averages and head-to-head records.
- Scheduled via Airflow with daily ingestion and weekly maintenance DAGs. Each sport has its own extractor (FBref, nba_api, nflreadpy) plugged into a shared interface — adding a new sport means writing one new extractor.

EDUCATION

University of Texas at Austin

2017

B.Sc.A. Human Development; Minor: Computer Science